MIL-C-60457 (MU)
27 October 1967
Supersedes
PA-PD-2690 Rev. 1
15 November 1965

MILITARY SPECIFICATION

CARTRIDGE, 105 MM, HE,
ROCKET ASSISTED, XM548,
BODY, MOTOR FOR

1. SCOPE

1.1 This specification covers the motor body for Cartridge, 105MM, HE, Rocket Assisted, XM548.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for.
MIL-I-6868 - Inspection Process-Magnetic Particles
MIL-H-6875 - Heat Treatment of Steels (Air Craft Practice, Process for)
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single or Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1315

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

DRAWING

ORDNANCE CORPS

9212147 - Box, Fiber, Inner, For Body Motor, XM548 (Metal Parts)
9212148 - Box, Fiber, Outer, For Body Motor, XM548 (Metal Parts)
9212398 - Body, Motor, XM548

PUBLICATIONS

ORDNANCE CORPS

ET-9212398 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing Materials

ASTM B 328-60 - Density and Interconnected Porosity of Sintered Powder Metal Structural Parts and Oil-Impregnated Bearings.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.).

3. REQUIREMENTS

3.1 Material.-Materials shall comply with the requirements specified on the applicable drawing and all referenced specifications.

3.2 Parts.-The parts shall comply with all requirements specified in the applicable drawings and specifications.

3.3 Banding

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.3.1 Band seat and band blank.-Prior to the application of the rotating band blank, the band seat and band blank shall be clean and free of oil, grease, dirt, rust, excess wax and other foreign material. Shot or sand blasting shall not be used for cleaning.

3.3.2 Band application.-The rotating band shall be applied in the cold state only. The band blank shall be carefully pressed into the band seat after final heat treatment and hydrostatic test of the motor body in such a manner as to prevent distortion of the adjacent walls and eliminate gap between the band and band seat. The rotating band shall make contact with the sidewalls of the band seat for a full 360 degrees and a minimum of 50 percent of the sidewall area. Distortion of the sidewalls shall be considered not to have taken place if the thread counterbore diameter remains within specified drawing tolerances. Once an acceptable banding pressure has been established, that pressure shall be maintained within plus or minus 25 PSI.

3.3.3 Density. - The density of the band, when pressed on the motor body, shall not exceed 6.45 grams per cubic centimeter.

3.4 Heat treatment.-The motor body shall be subjected to heat treatment in accordance with specification MIL-H-6875 to attain the required mechanical properties specified on the applicable drawing except that the body shall not be tempered between 500° F. and 700° F. All other deviations shall be subject to approval by the contracting officer.

3.5 Hydrostatic test.-The motor bodies shall withstand an internal hydrostatic pressure in accordance with the requirements on the drawing without evidence of cracks, ruptures, leakage or flaws. There shall be no permanent deformation at the location indicated on the drawing.

3.6 Surface finish of exit cone.-There shall be no tool marks, gouges, scratches, or other surface imperfections in the exit cone of the motor body. The surface finish shall be as specified on the drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.7 Metal defects.-There shall be no cracks, seams, laminations and laps in the motor body as evidenced by magnetic particle inspection. There shall be no cracks in the rotating band when pressed on the motor body. Non-metallic inclusions will be permitted in the motor body as specified in 3.7.1.

3.7.1 Non-metallic inclusions.-Each non-metallic inclusion indication 1/16 inch and over in length shall be recorded. Indications shall be no less than 3/8 inches apart. Motor bodies shall conform to a maximum frequency rating of 0.80 and a maximum severity rating of 0.67. The frequency (numbers) and severity (size) of indications shall be calculated as follows:

3.7.1.1 Frequency:

3.7.1.1.1 The number of indications per motor body shall be totaled.

3.7.1.1.2 The total number of indications shall be divided by 132 square inches (frequency per motor body).

3.7.1.2 Severity:

3.7.1.2.1 The length of each indication shall be recorded.

3.7.1.2.2 The summation of severities for each motor body shall be computed by totalling the products of the number of indications times the appropriate progression factor listed in the following table:

LENGTH OF INDICATION INCH	PROGRESSION FACTORS FOR SEVERITY RATING
1/16 to 1/8 inch	0.5
Over 1/8 to 1/4 inch	1
Over 1/4 to 1/2 inch	2
Over 1/2 to 3/4 inch	4
Over 3/4 to 1 inch	8
Over 1 to 1 1/2 inch	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.7.1.2.3 The summation of severities for each motor body is divided by 132 square inches (severity per motor body).

3.8 Serialization.-Each motor body shall be serialized for purposes of identification.

3.9 Threads.-The threads shall be undamaged for the entire length of the thread.

3.10 Proving ground.-The banded motor body when fired at a proving ground in a M103 or M137 cannon shall not collapse, deform or break up in the gun bore or in flight when fired using a propellant charge adjusted to produce 40,000 p.s.i. chamber pressure.

3.11 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.12 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of cracks, scratches, burrs, dents, gouges, holes, porous areas, distortions sharp edges, unblended radii, surface defects or any surface discontinuities. The item of shipment shall be free from dirt, chips, grease, corrosion, and other foreign material. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. Exterior surface coatings shall be continuous except for a few scratches not exposing base metal.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals, all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection.

4.2.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 35 motor body assemblies painted, 15 motor body assemblies unpainted and 10 complete sets of parts unpainted (i.e., 10 unbanded motor bodies and 10 rotating band blanks) which have been produced by the contractor or

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/certificates of conformance for materials, (see 4.1.1.e). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portions thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. - Assemblies and components will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 Lot formation. - The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantityof units from which the sample was drawn and shall not be construed to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specification, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of banded motor bodies shall conisst of not more than one mill heat of steel and not more than one lot of rotating band blanks.

4.3.1.1 Heat treat batch - A heat treat batch shall consist of motor bodies from one mill heat of steel which have been heat treated in the same final heat treating equipment at the same time, in one unchanged and continuous process. After final heat treatment, motor bodies shall remain grouped together for identification purposes until such time as the results of the mechanical property tests indicate compliance with the drawing requirements.

4.3.2 Examinations. - Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard Mil-Std-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard Mil-Std-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method shall be used for all of the Classification of defects listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.1 Body, Motor, prior to banding and after hydrostatic test (see dwg. 9212398).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
	1. Hyrostatic test stamp missing	Visual	01001
Major:	AQL 1.00 percent		
	101. Profile of rotating band seat	Gage	01002
	102. Foreign material on band seat	Visual	01003
	103. Diameter of rotating band seat	Gage	01004
	104. Band seat radius missing	Visual	01005
	105. Concentricity of rotating band seat with outside diameter	Gage	01006
	106. Angle of side wall of rotating band seat	Gage	01007
	107. Scores on band seat missing	Visual	01008
Minor:	AQL 1.50 percent		
	201. Distance threaded end surface to band seat	Gage	01009
	202. Max. flat on band seat side wall	Gage	01010
	203. Evidence of poor workmanship	Visual	01011

4.3.2.2 Body Motor, prior to banding and after hydrostatic test (see dwg. 9212398).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 100 percent		
	101. Magnetic Particle test stamp missing	Visual	02001
Minor: None defined.			

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.3 Body, Motor after banding (see dwg. 9212398).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 2.50 percent		
101.	Distance threaded end surface to 3.154 basic diameter, max.....	Gage	03001
102.	Distance threaded end surface to 3.764 basic diameter, min....	Gage	03002
103.	Distance threaded end surface to 4.030 basic diameter, min....	Gage	03003
104.	Distance threaded end surface to 3.420 basic diameter, max....	Gage	03004
105.	Concentricity of 3.764 basic diameter with outside diameter..	Gage	03005
106.	Concentricity of 3.154 basic diameter with inside diameter...	Gage	03006
107.	Concentricity of 4.030 basic diameter with outside diameter..	Gage	03007
108.	Concentricity of 3.420 basic diameter with inside diameter...	Gage	03008
109.	Diameter of cavity adjacent to large end of exit cone, max.....	Gage	03009
110.	Distance to large basic diameter in exit cone.....	Gage	03010
111.	Small diameter of exit cone.....	Gage	03011
112.	Distance to small basic diameter in exit cone.....	Gage	03012
113.	Concentricity of small diameter of exit cone with adjacent cavity.....	Gage	03013
114.	Diameter of thread undercut, max.....	Gage	03014
115.	Diameter of cavity adjacent to small end of exit cone, max.....	Gage	03015
116.	Radius in thread undercut missing.....	Visual	03016
117.	Shoulder at small end of exit cone perpendicular to adjacent cavity.....	Gage	03017
118.	Fillet radius at bottom of cavity adjacent to small end of exit cone missing.....	Visual	03018
119.	Maximum flat on exit cone.....	Gage	03019
120.	Length from threaded end surface to thread undercut.....	Gage	03020

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

121.	Width of thread undercut, max...Gage	03021
122.	Concentricity of thread counterbore with pitch diameter.....Gage	03022
123.	Wall thickness of spherical bulkhead.....Gage	03023
124.	Concentricity of inside diameter of large cavity with pitch diameter of thread...Gage	03024
125.	Inside diameter of large cavity, max.....Gage	03025
126.	Depth of cavity adjacent to small end of exit cone, min.....Gage	03026
127.	Concentricity of thread with outside diameter.....Gage	03027
128.	Surface finish of exit cone improper.....Visual	03028
129.	Surface finish of cavity adjacent to large end of exit cone improper.....Visual	03029
130.	Surface finish of thread counterbore improper.....Visual	03030
131.	Surface finish of threaded end improper.....Visual	03031
132.	Fillet radius at bottom of cavity adjacent to large end of exit cone missing.....Visual	03032
133.	Grooves in projection missing...Visual	03033
134.	Surface finish of inside spherical surface of large cavity.....Visual	03034
135.	Profile of rotating band.....Gage	03035
136.	Outside diameter of band.....Gage	03036
137.	Outside diameter of motor body..Gage	03037
138.	Threaded end surface perpendicular to outside diameter.....Gage	03038
139.	Pitch diameter of thread.....Gage	03039
140.	Diameter of thread counterbore..Gage	03040
Minor:	AQL 6.50 percent	
201.	Distance from threaded end surface to 2.25 basic diameter..Gage	03041
202.	Distance threaded end surface to spot face.....Gage	03042
203.	Perpendicularity of spot face with inside diameter.....Gage	03043
204.	Concentricity of cavity adjacent to large end of exit cone with cavity adjacent to small end of exit cone.....Gage	03044

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

205.	Shoulder adjacent to large end of exit cone perpendicular to adjacent cavity.....Gage	03045
206.	Diameter of cavity adjacent to large end of exit cone, min.....Gage	03046
207.	Distance from undercut in projection to shoulder adjacent to large end of exit cone.....Gage	03047
208.	Width of under cut in projection.....Gage	03048
209.	Spot face diameter, min.....Gage	03049
210.	Outside diameter of projection..Gage	03050
211.	Overall length, max.....Gage	03051
212.	Length of thread counterbore....Gage	03052
213.	Inside diameter of cavity in extreme end of projection.....Gage	03053
214.	Diameter of undercut in projection.....Gage	03054
215.	Distance from projection end surface to shoulder adjacent to large end of exit cone.....Gage	03055
216.	Chamfer on thread counterbore missing.....Visual	03056
217.	Chamfer behind undercut in projection missing.....Visual	03057
218.	Concentricity of cavity adjacent to small end of exit cone with pitch diameter of thread.....Gage	03058
219.	Blend radius at junction of projection outside diameter and external spherical surface missing.....Visual	03059
220.	Blend radius in large cavity at junction of inside diameter and tapered section missing.....Visual	03060
221.	Minor diameter of thread max....Gage	03061
222.	Evidence of poor workmanship....Visual	03062
223.	Concentricity of large end of exit cone with adjacent cavity..Gage	03063
224.	Spherical radius in bottom of large cavity.....Gage	03064
225.	Distance threaded end surface to end of rotating band.....Gage	03065
226.	Distance threaded end surface to 3.154 basic diameter, min....Gage	03066
227.	Distance threaded end surface to 3.764 basic diameter, max.....Gage	03067
228.	Distance threaded end surface to 4.030 basic diameter, max.....Gage	03068

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

229.	Distance threaded end surface to 3.420 basic diameter, min....Gage	03069
230.	Inside diameter of large cavity, min.....Gage	03070
231.	Depth of cavity adjacent to small end of exit cone, max.....Gage	03071
232.	Diameter of cavity adjacent to small end of exit cone, min.....Gage	03072

4.3.2.4 Body, Motor after banding (see dwg. 9212398).

Categories	Defects	Method of Inpsection	Code No.
Critical:	None defined.		
Major:	AQL 100 percent		
101.	Magnetic particle test stamp missing.....Visual		04001
Minor:	None defined.		

4.3.2.5 Body, Motor after painting (see 9212398).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Marking missing, misleading incomplete or illelible.....Visual		05001
202.	Protective finish incomplete or damaged exposing base metal.....Visual		05002
203.	Evidence of poor workmanship....Visual		05003

4.3.2.6 Blank, Rotating Band (see dwg. 9212400).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Width of band.....Gage		06001
102.	Excessive wax on band blank.....Visual		06002

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

Minor:	AQL 1.50 percent		
201.	Inside diameter max.	Gage	06003
202.	Thickness, max.	Gage	06004
203.	Evidence of poor workmanship.	Visual	06005

4.3.2.7 Box, Fiber, Outer, for Body Motor, XM548 (Metal Parts) (see dwg. 9212148).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Contents loose.	Visual/Manual	07001
102.	Air in bag not exhausted.	Visual	07002
103.	Bag improperly sealed.	Visual	07003
Minor:	AQL 1.50 percent		
201.	Carton seal loose.	Visual/Manual	07004
202.	ICC marking missing.	Visual	07005
203.	Outer box improperly sealed.	Visual	07006
204.	Marking improper, incomplete or illegible.	Visual	07007

4.3.3 **Testing**

4.3.3.1 **Rotating band**

4.3.3.1.1 **Tightness** - Code No. 03073 - Major defect - Two banded motor bodies from every 1000 or less motors submitted shall be selected for this test. If either of the sample motor bodies fail to meet the requirements of 3.3.2, the 1000 or less motor bodies from which the sample was selected, shall be rejected. When applicable, the same motor bodies used in 4.3.3.2.1.2 may be used in conducting this test.

4.3.3.1.2 Density. - Code No. 03074 - Major defect - After examination for band tightness as specified in 4.3.3.1.1, one of the segments from each sample motor body shall provide a density test specimen. If the density of either specimen exceeds the requirement of 3.3.3, the 1000 or less motor bodies represented by the sample shall be rejected.

4.3.3.1.3 Application. - Major defect - Code No. 03075 - The contractor shall provide adequate control to assure that banding pressure does not vary in excess of the requirements of 3.3.2. If during the banding operation, the pressure varies beyond the specified requirement, the motor shall be rejected and removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.3.2 Motor Body

4.3.3.2.1 Mechanical Properties

4.3.3.2.1.1 Hardness. - Code No. 01012 - Each motor body from a heat treated lot shall be serialized and subjected to a hardness test on the surface indicated. Hardness readings shall be recorded for each serialized body. The test shall be conducted in accordance with Federal Test Method Standard No. 151, Method 243.1. The hardness readings thus obtained shall be used as the basis for selecting the tension test samples of paragraph 4.3.3.2.1.2.

4.3.3.2.1.2 Tensile test (see 4.3.3.2.1.1) - Major defect - Code No. 01015 - Two motor bodies from every heat treat batch or motor body lot whichever is less shall be selected for this test (one motor body from the highest hardness reading and one motor body from the lowest hardness reading). From each of these motor bodies, two longitudinal test specimens taken 180 degrees apart, shall be tested to provide yield strength and elongation data. Testing shall be conducted in accordance with Federal Test Method Standard No. 151, Method 211.1 using an R5 test specimen. If any specimen fails to meet the requirements of the drawing, the heat treated motor bodies shall be rejected subject to retest as specified in 4.3.3.2.1.2.1. Ultimate strength, reduction in area, and hardness of the test specimen shall be recorded for informational purposes only.

4.3.3.2.1.2.1 Retest. - When one or more tensile test specimens fail to comply with the yield strength or elongation requirements of the drawing, new samples of a higher or lower hardness values or both, as applicable, shall be selected to establish an acceptable hardness range for the motor bodies. When the acceptable hardness range has been established, all motor bodies which fall outside the range shall be rejected subject to review by the Government.

4.3.3.2.2 Hydrostatic test (see 3.5) - Critical defect - Code No. 01013 - Each motor body (unbanded) shall be hydrostatically tested in accordance with 4.4.2 using the test equipment of 4.3.4. If any motor body fails to meet the requirements of 3.5, the motor body shall be classed defective and removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.3.2.3 Magnetic particle - Major defect - Each motor body shall be magnetic particle inspected before banding, Code No. 01014, and after banding, Code No. 03076. If any motor body fails to meet the requirement of 3.7 before or after banding, the motor body shall be rejected and removed from the lot. The test shall be conducted in accordance with Specification MIL-I-6868.

4.3.3.2.4 Proving Ground tests. - The following tests shall be performed as specified in 4.4.3. Motor bodies for these tests shall be forwarded to a proving ground as specified by the contracting officer.

4.3.3.2.4.1 Security - Critical defect - Code No. 05004 - Twenty motor bodies from the first article sample and ten motor bodies randomly selected from each production lot shall be subjected to this test. If any of the sample motor bodies fail to meet the requirements of 3.10, the lot shall be rejected. Range and deflection shall be recorded for information only.

4.3.4 Inspection Equipment - Equipment Tabulation Number ET-9212398 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection equipment. The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4 Test Methods and Procedures.

4.4.1 Rotating Band.

4.4.1.1 Band tightness. - Four 90 degree segments shall be cut from the open end of the motor body. A specimen approximately 1 1/4 inches long shall be cut from each segment and examined for the requirements of 3.3.2. Each specimen shall be polished, etched and examined using the equipment specified on the Inspection Equipment List.

4.4.1.2 Density test. - The density of the band shall be determined using the procedure described in ASTM Test Method B 328-60. A specimen one half inch minimum length and full width of the band shall be removed from one of the segments used in the band tightness test of 4.4.1.1.

4.4.2 Hydrostatic test (see 6.4). - The motor body shall be mounted in a suitable test fixture and tested in accordance with the drawing requirements. The motor body shall be temporarily marked and measurements of the outside diameter shall be taken before and after the test at the marked location to determine permanent deformation in the area indicated on the drawing. The motor body shall be cleaned and dried immediately after the test to prevent corrosion.

4.4.3 Proving ground tests.

4.4.3.1 Security. - The motor bodies supplied in accordance with 4.3.3.2.4.1 shall be inert loaded and assembled to an inert warhead and an inert fuze and fired in a gun using a propellant charge adjusted to produce 40,000 p.s.i. chamber pressure in a M103 or M137 cannon. Motor bodies shall be recovered and examined for conformance with 3.10 and the results recorded. Any change in dimensions exceeding .001 inch will be considered as deformation.

4.4.4 Magnetic Particle Inspection. - After hydrostatic testing, the entire unbanded motor body shall be magnetic particle inspected on the internal and external surfaces. The band shall be applied and a second magnetic particle inspection performed on the motor body. The motor body internal and external surfaces, (including the rotating band) shall be investigated for cracks for distance of two (2) inches, minimum from the threaded end. Magnetic particle inspection shall be in accordance with MIL-I-6868 using the indirect, continuous, wet fluorescent method. Kerosene shall not be used for carrying the magnetic particle substance when inspecting the motor body after banding.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.5 Band application. - A suitable warning device shall be assembled to each banding machine to assure pressure is maintained within specified tolerances. Equipment shall be as specified on the Inspection Equipment List.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level C - Packaging of the motor body shall be in accordance with drawing 9212147.

5.2 Packing

5.2.1 Level C - Packing of the motor body shall be in accordance with drawing 9212148.

5.3 Marking - Marking shall be in accordance with the applicable packaging and packing drawings.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard Mil-Std-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accor-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

dance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Deformation. - Deformation is defined as an enlargement of the dimension in excess of .001 inch.

6.5 Submission of First Article Samples. - Instructions as to location of first article samples shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall adivse Picatinny Arsenal and instructions issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

6.6 Submission of test data. - In addition to the normal distribution of records, one copy of all test data shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND4 and SMUPA-DS2
Dover, New Jersey

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project No. 1315-A-572

GPO 303-223/Z 661

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.